MODULE 5: COMPUTATIONAL CHEMISTRY: BASIC TO APPLICATIONS

(CO: 4)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACHY101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 10 Topics): (1) Introduction to Quantum mechanical calculations, (2) Foundations of computational chemistry, (3) Molecular orbital theory: Atomic orbitals and molecular orbitals, (4) Orbital interactions and chemical bonding, (5) Potential energy curve/surface (PES), (6) Bio-molecular potentials, (7) Geometry optimization, (8) Vibrational frequency calculations, (9) Reaction mechanism elucidation, and (10) Application of AI to screen semiconductors.

1 VIT FAT Exam Blueprint & Master Formula Sheet

Module 5 carries **15 to 20 marks** in BACHY101 FAT:

1. **Question Type A (10 Marks):** Molecular Orbital Theory and LCAO principle: Energy level diagrams for homonuclear (N_2, O_2) and heteronuclear (CO, HF) diatomics, bond order, and magnetic property interpretation.
2. **Question Type B (10 Marks):** Potential Energy Surface (PES): Local vs global minima, transition states (saddle point), geometry optimization gradient methods, and vibrational frequency analysis (imaginary frequency for transition state).
3. **Question Type C (10 Marks):** Computational screening: Lennard-Jones biomolecular potentials and Machine Learning / AI algorithms for predicting semiconductor bandgaps and solar cell efficiencies.

Concept	Governing Equation / Algorithm	VIT Exam Key Insight
Time-Independent Schrodinger Eq	$\hat{H}\psi = E\psi$, $\hat{H} = -\frac{h^2}{8\pi^2 m}\nabla^2 + V$	Basis of all electronic structure methods (DFT, Hartree-Fock).
LCAO Approximation	$\psi_{MO} = c_1\varphi_1 + c_2\varphi_2$	Bonding MO: $\psi_+ = \frac{\varphi_1+\varphi_2}{\sqrt{2}}$; Antibonding: $\psi_- = \frac{\varphi_1-\varphi_2}{\sqrt{2}}$.
Bond Order Formula	Bond Order $= \frac{N_b - N_a}{2}$	Paramagnetism of O_2 explained by 2 unpaired electrons in $\pi^*_{2p_x} = \pi^*_{2p_y}$.
Lennard-Jones 12-6 Potential	$V_{LJ(r)} = 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$	r^{-12} is short-range Pauli repulsion; r^{-6} is London dispersion attraction.
Transition State Condition	Hessian $H_{ij} = \frac{\partial^2 E}{\partial x_i \partial x_j}$ has exactly ONE negative eigenvalue	One imaginary vibrational frequency corresponds to reaction coordinate.

2 Molecular Orbital Theory & Orbital Interactions

2.1 Principles of LCAO (Linear Combination of Atomic Orbitals)

Molecular orbitals (MOs) are formed by linear superposition of atomic orbitals (AOs) when:

1. The overlapping AOs have comparable energies.
2. The AOs possess compatible spatial symmetry with respect to the internuclear axis (z-axis).
3. Significant spatial overlap occurs ($S_{12} = \int \varphi_1^* \varphi_2 d\tau > 0$).

For two identical atomic orbitals φ_1 and φ_2:

- **Bonding Molecular Orbital (ψ_b):**

$$\psi_b = \frac{1}{\sqrt{2(1+S)}}(\varphi_1 + \varphi_2)$$

Constructive interference increases electron probability density in the internuclear region, stabilizing the molecule ($E_b < E_{\text{AO}}$).

- **Antibonding Molecular Orbital (ψ_a):**

$$\psi_a = \frac{1}{\sqrt{2(1-S)}}(\varphi_1 - \varphi_2)$$

Destructive interference creates a nodal plane perpendicular to the internuclear axis ($E_a > E_{\text{AO}}$).

2.2 Molecular Orbital Diagrams (O_2 and N_2)

- **For N_2 ($14e^-$) with $s-p$ mixing:**

$$\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\left(\pi_{2p_x}^2 = \pi_{2p_y}^2\right)\sigma_{2p_z}^2$$

Bond Order $= \frac{10-4}{2} = 3$ (Diamagnetic).

- **For O_2 ($16e^-$) with negligible $s-p$ mixing:**

$$\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2p_z}^2\left(\pi_{2p_x}^2 = \pi_{2p_y}^2\right)\left(\pi_{2p_x}^{*1} = \pi_{2p_y}^{*1}\right)$$

Bond Order $= \frac{10-6}{2} = 2$. MOT brilliantly predicts **paramagnetism** due to two degenerate half-filled antibonding orbitals (π^*), which valence bond theory failed to explain!

3 Potential Energy Surfaces (PES) & Reaction Mechanisms

3.1 Topography of Potential Energy Surfaces

A Potential Energy Surface expresses the molecular electronic energy $E(\boldsymbol{R})$ as a function of nuclear coordinates:

- **Local / Global Minima:** Stationary points where the gradient vanishes ($\nabla E = 0$) and all eigenvalues of the Hessian matrix ($\partial^2 \frac{E}{\partial} x_i \partial x_j$) are strictly positive. Represents stable reactants, intermediates, or products.
- **Transition State (First-Order Saddle Point):** Stationary point where $\nabla E = 0$, but the Hessian matrix possesses **exactly one negative eigenvalue** (corresponding to an imaginary vibrational frequency along the intrinsic reaction coordinate). All other eigenvalues are positive.

3.2 Geometry Optimization & Vibrational Frequencies

- **Gradient Descent / Quasi-Newton Methods (BFGS):** Iteratively adjust nuclear coordinates along the force vector $\boldsymbol{F} = -\nabla E$ until maximum force $|\boldsymbol{F}| < 10^{-4}$ Hartree/Bohr.
- **Vibrational Frequency Analysis:** The second derivative matrix yields vibrational normal modes. A true minimum has all real, positive vibrational frequencies. A transition state has exactly one imaginary frequency ($\nu < 0\text{cm}^{-1}$), confirming the barrier summit.

4 Biomolecular Potentials & AI Screening of Materials

4.1 Molecular Mechanics Force Fields

For large biomolecules (proteins, DNA) where quantum mechanics is too computationally demanding, empirical force fields model the potential energy:

$$V_{\text{total}} = \sum_{\text{bonds}} k_b(r-r_0)^2 + \sum_{\text{angles}} k_\theta(\theta-\theta_0)^2 + \sum_{\text{dihedrals}} V_n[1+\cos(n\varphi-\gamma)] + \sum_{\text{non-bonded}}\left[4\varepsilon\left(\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right) + \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}\right]$$

4.2 AI & Machine Learning for Screening Semiconductors

Modern materials discovery utilizes graph neural networks (GNNs) and density functional theory (DFT) datasets:

1. **Crystal Graph Convolutional Networks (CGCNN):** Represent crystals as graphs where nodes are atoms and edges are bond distances.
2. **High-Throughput Property Prediction:** Trains on DFT databases (Materials Project) to predict bandgaps (E_g), electron mobilities (μ_e), and formation enthalpies within milliseconds instead of days.
3. **Application:** Accelerates discovery of lead-free halide perovskites ($Cs_2AgBiBr_6$) and wide-bandgap semiconductors for high-efficiency solar cells and EV power inverters.